

Principled loss functions for continuous-time learning

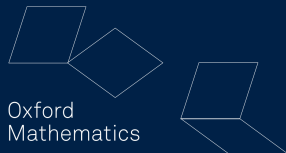


Mathematical
Institute

Haoping Qiu

*Mathematical Institute
University of Oxford*

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Motivation

Path closeness in continuous time

Path-to-path models produce whole trajectories. Training therefore needs a precise rule for comparing two paths.

Project question

The project studies definitions of path closeness and how each definition changes the learned path or operator. Suppose two predictions agree with a target at every recorded time:

$$\hat{Y}_A(t_j) = \hat{Y}_B(t_j) = Y(t_j), \quad j = 0, \dots, m.$$

Pointwise MSE gives both predictions zero loss. Their behaviour between observations can still differ, so

$$\int_0^T \|\hat{Y}_A(t) - Y(t)\|_2^2 dt \neq \int_0^T \|\hat{Y}_B(t) - Y(t)\|_2^2 dt.$$

Role of discretisation

Finite storage does not remove the issue. Weighting, interpolation and feature maps select the continuous error being approximated.

Path-to-path learning

A model maps one evolving object to another. Input path $X^{(i)} : [0, T] \rightarrow \mathbb{R}^{d_x}$ is an observed driver or stream, and predicted response is

$$\hat{Y}_\theta^{(i)} = \mathcal{F}_\theta(X^{(i)}).$$

Training uses paired input and target paths $(X^{(i)}, Y^{(i)})$ and selects

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(\hat{Y}_\theta^{(i)}, Y^{(i)}).$$

Discrepancy $\mathcal{L}$ decides which prediction errors training treats as large. Every experiment below changes $\mathcal{L}$ and holds everything else fixed.

Observation density and the sampled loss

Let observation times have density q on $[0, T]$ and let $e_\theta(t) = \hat{Y}_\theta(t) - Y(t)$. Then

$$\frac{1}{m} \sum_{j=1}^m \|e_\theta(t_j)\|_2^2 \longrightarrow \int_0^T \|e_\theta(t)\|_2^2 q(t) dt.$$

Trapezoidal elapsed-time weights instead approximate

$$J_2(\theta) = \int_0^T \|e_\theta(t)\|_2^2 dt.$$

Uniform observations

q is constant. MSE and J_2 differ mainly through endpoint weights.

Clustered observations

MSE rewards accuracy where points are dense.
 J_2 restores elapsed-time emphasis.

Order and integral norms

Every L^p discrepancy compares residual magnitude at aligned times, so it depends on the path only through how long the residual spends at each size. Any measure-preserving rearrangement of time leaves it unchanged:

$$\|f\|_{L^p} \text{ depends on } f \text{ only through } \mu_f(\lambda) = |\{t : |f(t)| > \lambda\}|.$$

Permuting when events happen is acceptable in a norm and disqualifying in a path-to-path loss, where order is the structure being learned.

Consequence for rough streams

A second-level perturbation can leave every coordinate value unchanged. Coordinate losses then respond by exactly zero, whatever their exponent or weighting.

What signatures add

Level k is an iterated integral over $t_1 < \dots < t_k$, so it is built from ordering rather than from magnitude at aligned times.

Losses

Four loss families

Loss	Definition	Emphasis
Pointwise MSE	$\frac{1}{m} \sum_{j=1}^m \ \hat{Y}(t_j) - Y(t_j)\ _2^2$	Each recorded point has equal weight
Elapsed-time J_2	$\int_0^T \ \hat{Y}(t) - Y(t)\ _2^2 dt$	Error accumulated over time
Sobolev H^1	$J_2 + \kappa \int_0^T \ \dot{\hat{Y}}(t) - \dot{Y}(t)\ _2^2 dt$	Values and local velocity for smooth paths
Signature loss	$\ \Phi(\hat{Y}) - \Phi(Y)\ _2^2$	Ordered increments and interactions

Comparison principle

Weight $\kappa > 0$ balances value against velocity error; Φ is a truncated signature feature map, defined on the signature slide. Train a matched model under each objective, then evaluate every fitted path under every metric.

Integral and Sobolev norms: [1]. Signatures and rough paths: [2,3,8]

For a path $Y : [a, b] \rightarrow \mathbb{R}^d$, its depth- M truncated signature is

$$S^{\leq M}(Y)_{a,b} = \left(1, \int_a^b dY, \int_{a < t_1 < t_2 < b} dY_{t_1} \otimes dY_{t_2}, \dots \right).$$

Level 1

Net displacement

Level 2

Ordered pair interactions and
signed area

Higher levels

Finer ordered interactions

Two constructions used in Experiment A

Global loss compares one depth-4 signature. Local loss compares depth-2 signatures over a fixed partition, with an anchor for starting position.

Signature foundations: Chen (1957) [2], Lyons, Caruana and Lévy (2007) [3], Chevyrev and Kormilitzin (2016) [8]. Project definitions: Notebook 05, Sections 5 and 6

Signature losses for training

Write $\Delta S^{(k)} = S^{(k)}(Z_{\hat{Y}}) - S^{(k)}(Z_Y)$, with Z the time-augmented output path and $d_Z = 3$. Each training loss is a choice of level weight and interval.

Variant	Form	Choice it encodes
Global	$\frac{1}{d_Y} \ \hat{Y}(0) - Y(0)\ _2^2 + \sum_{k=1}^4 d_Z^{-k} \ \Delta S^{(k)}\ _2^2$	Level weighted by coordinate count alone
Local	$\frac{1}{K} \sum_{b=1}^K \sum_{k=1}^2 d_Z^{-k} \ \Delta S_{I_b}^{(k)}\ _2^2, K = 10$	Features attached to coarse time location
Refined local	Same with $K = 100$ and factor $(K/K_0)^{2k}$	Homogeneity correction, so block count alone changes nothing

Anchor terms appear because signatures use increments and are blind to a constant shift. Level weighting is an open choice: signature theory fixes none of these, so levelwise value and gradient are audited before any comparison.

Signature discrepancies for measurement

Study C trains nothing, so the same features can be weighted by observed variation rather than by convenience.

Variant	Form	Choice it encodes
Variance normalised	$\sum_{k=1}^4 \ \Delta S^{(k)}\ _2^2 / (d_Z^k (q_k^2 + \delta))$	Levels balanced by Brownian variation q_k measured on training streams
Aligned and offset local	Depth two over 128 blocks of 16 increments, and the same shifted by 8	Isolates partition placement against the message
Response derived	$\sum_{k=1}^L r^k \ \Delta S^{(k)}\ _1, r \in \{0.5, 1, 2\}$	r bounds one-step response of a linear controlled system

Response weight r^k is not a convention. A level- k term of a linear controlled response carries k coefficient matrices, so bounding each by r bounds that term by r^k , and sweeping r asks how much high-level weight a message needs before it is visible.

Experimental design

Three studies

A. Fixed target

One analytic target path, fitted repeatedly. No input variation and no generalization, so a difference between fitted paths comes from capacity, optimizer or loss.

B. Learned operator

Many Brownian drivers with paired OU responses, held-out test paths. Adds generalization and causality to the same loss comparison.

C. Message sensitivity

No model and no training. Perturb a stream by a known amount and measure how far each discrepancy moves.

Reason for keeping C separate

In A and B a loss reaching a poor fit is ambiguous between a representation that ignores the feature and an optimizer that failed to minimize its own objective. Both happened: global-signature training in A finishes worse on its own discrepancy than a pointwise fit. Study C removes optimization entirely, so its answer describes the representation alone.

Experiment A: fixed target

Controlled Neural ODE fit

Fit one fixed analytic two-dimensional target path:

$$\dot{z}_\theta(t) = f_\theta(t, z_\theta(t)), \quad \hat{Y}_\theta(t) = g_\theta(z_\theta(t)).$$

- ▶ 64 target observations
- ▶ uniform or clustered observation times
- ▶ restricted or expressive model capacity
- ▶ 3 paired seeds and 10,000 Adam updates

Reason for the fixed target

Input variation and generalization are removed. Differences between fitted paths come from model capacity, optimization and training loss.

Matched comparison

Target, initialization, optimizer budget and model family remain fixed within each group. Only training loss changes. Evaluation uses a dense 1,025-point grid and common metrics.

Procedure: Notebook 05, Sections 3 to 9. Neural ODE model: Chen et al. (2018) [4]

Experiment B: Brownian driver to OU response

Brownian driver W determines an Ornstein–Uhlenbeck response

$$dY_t = -\lambda Y_t dt + \sigma dW_t, \quad Y_0 = 0.$$

A Neural CDE learns the whole map $W \mapsto Y$

$$dh_t = f_\theta(h_t) d\tilde{X}_t, \quad \hat{Y}_t = g_\theta(h_t).$$

Control $\tilde{X}_t = (t, W_t)$ is the driver augmented by time.

Matched study

- ▶ 512 train, 128 validation and 256 test paths
- ▶ 64 target times per response
- ▶ uniform or clustered target sampling
- ▶ 3 seeds per primary loss

Purpose

Test whether the sampling mechanism persists when one model learns across many input and response paths.

Procedure: Notebook 06, Sections 2 to 5. Neural CDE model: Kidger et al. (2020) [5]

Experiment B: procedure

Paired generation

One draw of standard-normal increments produces both the Brownian driver and, through Euler–Maruyama, its OU response. Target is therefore the response to *that* driver, not an independent OU path.

Causal model

One RK4 step per interval integrates $dh = f_\theta(h) dX$, so the output at time t depends only on driver increments before t .

Acceptance before comparison

Gates run before any loss comparison: the generator reproduces its own Euler recurrence exactly; perturbing the driver after the midpoint leaves earlier output unchanged and later output changed; every parameter receives finite gradient; data, initialization, target times and batch order all reproduce by fingerprint.

Matched runs

Within a seed and condition, every loss shares inputs, targets, initialization and batch order. Only the loss changes.

Procedure and acceptance: Notebook 06, Sections 2 to 6. Generator: `brownian_ou-pairs` in `src/pathloss/datasets.py`

Experiment C: rough-path streams

Elapsed-time weighting changes where coordinate error counts. It cannot measure information absent from coordinate values. Each supplied stream contains 2,048 step-two increments in four dimensions:

$$\mathbf{X}_j = (a_j, A_j), \quad a_j \in \mathbb{R}^4, \quad A_j \in \Lambda^2 \mathbb{R}^4.$$

First level a_j

Four coordinate increments. Their cumulative sum gives the visible piecewise path.

Second level A_j

Six Lévy-area coordinates. They record within-step order and affect rough differential equation responses.

Independent Gaussian coordinate increments do not determine the correct joint Lévy-area law. The supplied simulation includes both levels.

Data specification: Notebook 07, Sections 3 and 4. Brownian area simulation: Wiktorsson (2001) [6], Foster and Habermann (2023) [7]

Experiment C: procedure

Two readings

Paired: compare each modified stream against its own unmodified original, and normalise by the median distance between two independent streams, so a value of one means the message is as large as ordinary Brownian variation.

Detection: hand a classifier one stream and ask whether it is modified, with originals and modifications drawn from disjoint underlying paths.

Amplitude zero is a null control: it must leave every stored coordinate unchanged bit for bit and every discrepancy at zero.

Controls fixed in advance

Amplitude grid, message support, discrepancy definitions and split are frozen before results. A second 10,000-stream ensemble stays untouched until every choice is fixed, then repeats the transitions.

Independent verification

RoughPy reproduces the step-two signatures of original and modified streams to 10^{-9} , so the perturbation is checked against a library outside this project.

Message construction

A 16-increment block at the stream midpoint receives a message of normalized size ρ .

$$b_r = \rho \sigma_1 s_r u, \quad B_r = \rho \sigma_2 s_r v, \quad \rho \in \left\{0, \frac{1}{16}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4\right\}.$$

Scales σ_1, σ_2 are training-set standard deviations of first- and second-level increments. Fixed directions $u \in \mathbb{R}^4$ and $v \in \Lambda^2 \mathbb{R}^4$ select one coordinate at each level, and s_r is the template below.

Message	Modified component	Intended probe
Increment	a_j only	Coordinate displacement
Area	A_j only	Hidden second-level information
Combined	a_j and A_j	Mixed-level sensitivity
Net template	$s_r = 1$	Basic response to accumulated change
Balanced template	Eight +1, then eight -1	Local order with zero direct sum

Path view	Distances and features	Information measured
Coordinate	MSE, J_2 , L^1 , maximum error, direct increment	First-level coordinate motion
Rough path	Direct area, global depth-4 signature, local depth-2 signatures	Second-level area and order
Derived	Truncated signature kernel, response-derived signatures	Reweighted signature information

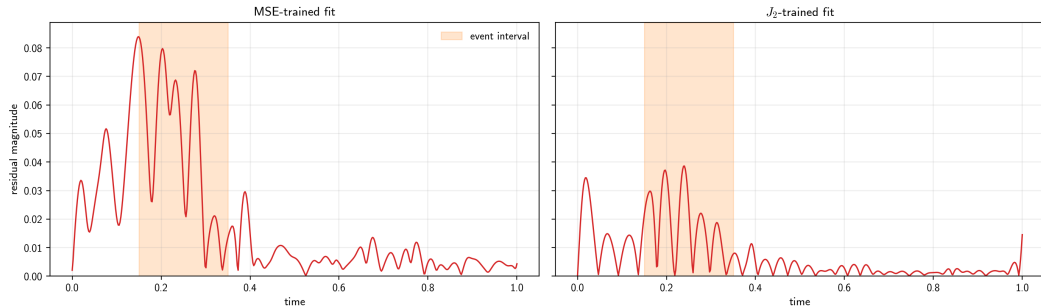
Two readings of sensitivity

Direct comparison measures each modified stream against its exact original. A 1-nearest-neighbour classifier tests whether one distance separates the two populations.

Metrics: Notebook 07, Sections 7 and 8. Signature kernels: Salvi et al. (2021) [9]. Response representations: Ferrucci, Perrée and Lyons (2026) [10]

Results

Experiment A: sampling density



Paired result

Under clustered sampling, J_2 lowers dense elapsed-time error in all 6 comparisons. Ratio-of-means reductions are **68.4%** for restricted models and **40.8%** for expressive models. Uniform sampling gives a 3 to 3 split.

Experiment A: derivative supervision

H^1 result

H^1 training gives the best general errors in all 6 groups at update 1,000 and 5 of 6 groups at update 10,000. It directly penalizes velocity error, so it resolves the target's short oscillatory burst efficiently.

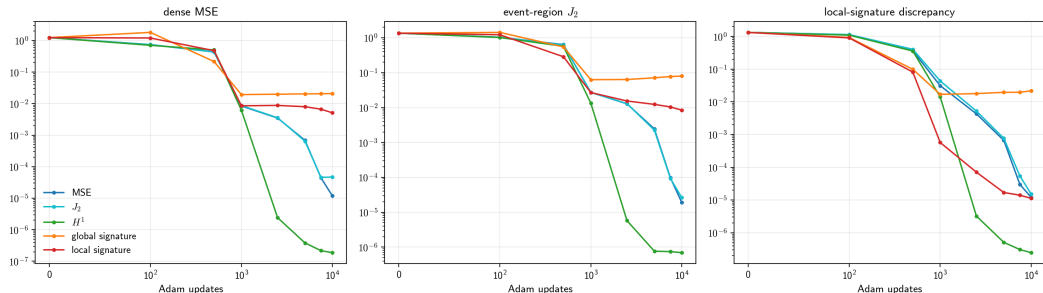
Interpretation limit

This target is analytic and its derivative is known. Brownian paths are almost surely outside H^1 , so the same supervision is unavailable for rough stream tasks.

Smooth targets can reward derivative-aware losses more than generic structural summaries. Signature losses are compared against this control rather than against MSE alone.

Results: Notebook 05, Sections 17 and 21. Sobolev viewpoint: Ramsay and Silverman (2005) [1]

Experiment A: learning dynamics



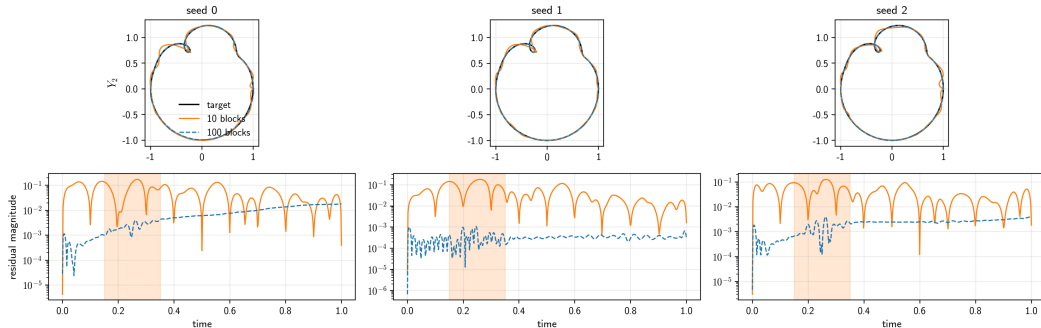
Early advantage is objective-specific

At update 500 global-signature training leads on its own discrepancy in all 6 groups and local-signature training in 5 of 6. Neither leads on event-region error, where pointwise and H^1 losses are ahead in 5 of 6.

Terminal fits alone would hide this

Signature advantage is transient. Reporting only update 10,000 would record a null where an ordering exists earlier in training.

Experiment A: truncation and partition



Global depth 4

At the final update, pointwise-trained fits have lower global-signature error than the global-signature-trained fit in all 6 groups. Optimization limits this arm.

Local depth 2

Refining the partition from 10 to 100 intervals lowers MSE, J_2 , H^1 , maximum error and both local-signature discrepancies in all 6 groups.

Experiment B: operator learning

Uniform target times

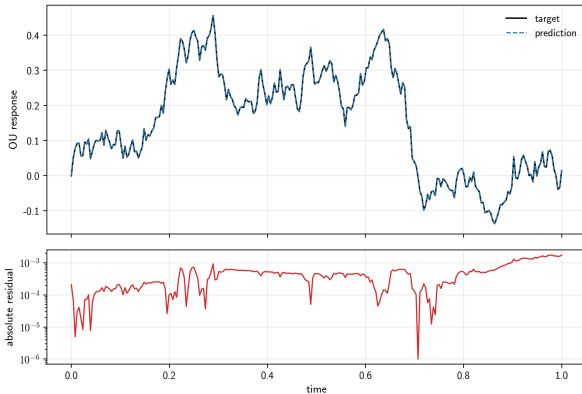
MSE and J_2 give nearly identical held-out error. Equal point weights already approximate elapsed-time weighting.

Clustered target times

J_2 lowers dense test J_2 for all 3 paired seeds. Its across-seed mean is **51.7% lower** than MSE training. Dense test MSE falls by 51.6%.

Signature losses

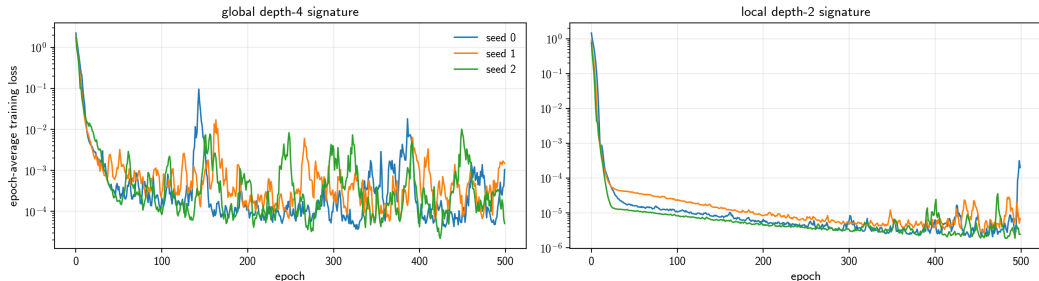
Global-signature training is unstable across all 3 seeds. Local-signature training improves its own metric in 2 of 3 seeds, with one unstable run.



Clustered J_2 fit, seed 1, held-out test path 0. Residual stays below 2×10^{-3} , so loss comparison concerns residual scale rather than shape.

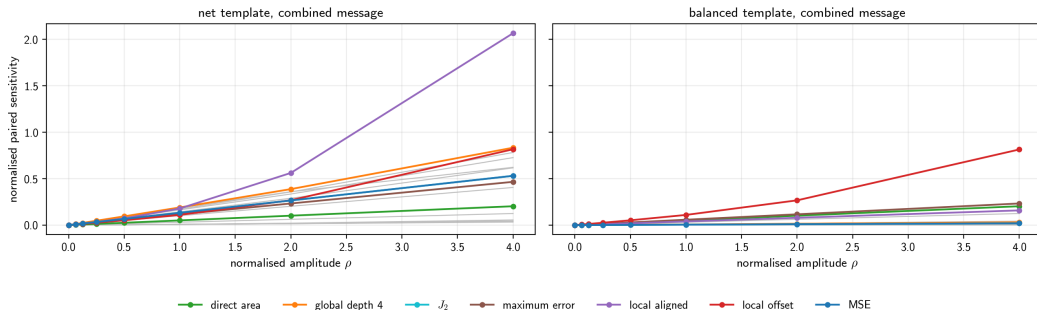
Results and stored predictions: Notebook 06, Sections 10 to 12

Experiment B: optimizer stability



Both signature objectives reach much lower training loss early, then rise. Final-to-best training-loss ratios are 28.4, 28.5 and 2.32 for the global loss. Terminal fits therefore record optimizer behaviour rather than a path the geometry prefers, so every signature claim in this project is stated at a fixed budget.

Experiment C: message templates



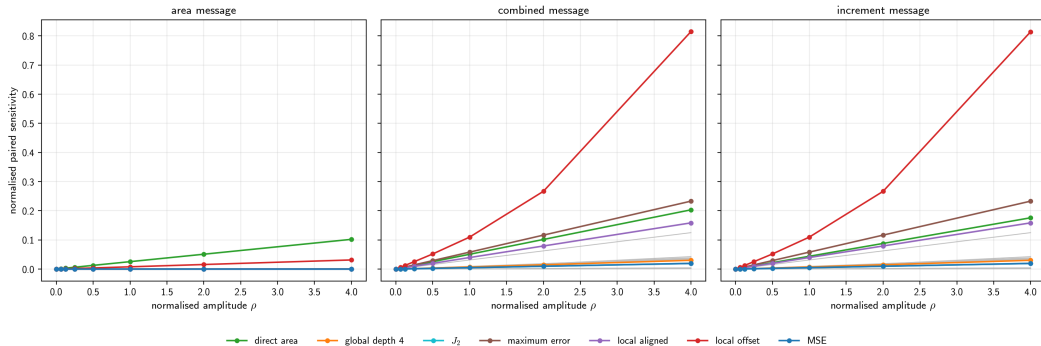
Net template

Inserted increments accumulate, so endpoint displacement alone separates the classes and every discrepancy responds.

Balanced template

Direct increments sum to zero over the block. Response now depends on whether a representation retains local order, which is the discriminating case.

Experiment C: balanced messages



Area only

Coordinate losses remain zero because a_j is unchanged. Area and signature features respond.

Combined

Offset local signatures amplify a message whose direct increments cancel over the full block.

Increment only

Pointwise losses respond to the local excursion. Signature response depends on partition placement.

Observed values and figure code: Notebook 07, Section 10. Stored main-study paired sensitivities

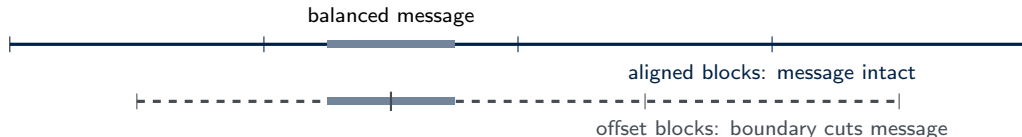
Experiment C: partition alignment

Aligned local signature

A comparison interval contains the full balanced message. Added level-one increments and direct level-two areas sum to zero, so depth two can cancel.

Offset local signature

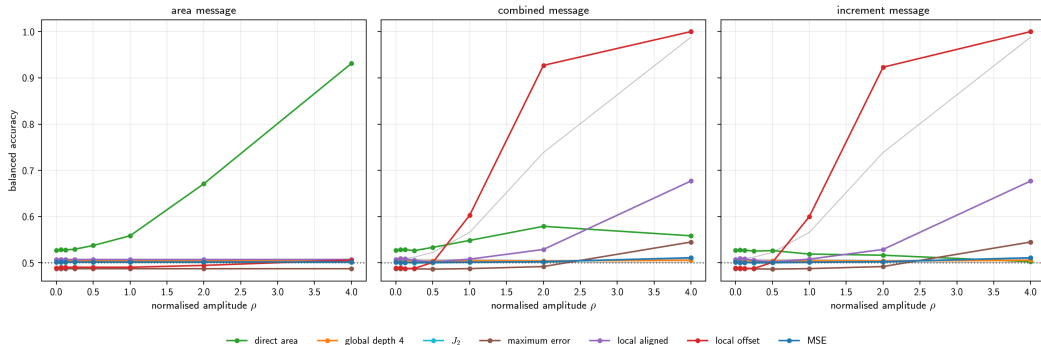
A boundary cuts through the message. Partial sums no longer cancel inside each interval, creating a deterministic depth-two response.



Partition choice is part of the loss definition.

Construction, derivation and observed sensitivity: Notebook 07, Sections 5, 7 and 10

Experiment C: single-stream detection



A 1-nearest-neighbour classifier uses one loss-derived distance and asks whether modified streams separate from original streams; the dotted line marks 50% chance accuracy. For balanced increment and combined messages, direct increment features detect from $\rho = 0.5$, offset local signatures from $\rho = 1$, aligned local signatures from $\rho = 2$, and maximum error from $\rho = 4$. Every transition reproduces on the independent confirmation ensemble.

Conclusions

Evidence	Main result	Meaning of closeness
Sampling limit	MSE targets a density-weighted integral	Observation frequency
Fixed path and OU	J_2 corrects clustered bias	Elapsed time
Smooth fixed path	H^1 resolves a short oscillatory burst	Values and derivatives
Brownian messages	Signature features detect area and order	Ordered multilevel structure

Project conclusion

Two paths count as close only relative to structure relevant to the task. Observation density, regularity, scale, signature truncation, localization and optimization can each change the comparison.

Evidence synthesis: notebooks 01, 05, 06 and 07

Scope and continuation

Supported conclusions

- ▶ Use elapsed-time weights under irregular sampling.
- ▶ Use derivative penalties when smooth derivatives are meaningful.
- ▶ Use signature features when order or area matters.
- ▶ Treat signature partition and scaling as model choices.

Scope

One fixed target, one operator, one Brownian law, finite training budgets and three seeds. Signature training remains sensitive to optimization.

Natural continuation

Hybrid losses, multiscale local signatures, learned path representations and broader stream operators.

Scope and recorded decisions: notebooks 05 to 07 and logbook entry 8 September 2026

Appendix: Experiment A comparison matrix

Training objectives

- ▶ MSE and elapsed-time J_2
- ▶ H^1
- ▶ global depth-4 signature
- ▶ local depth-2 signature over 10 blocks
- ▶ refined local depth-2 signature over 100 blocks

Evaluation metrics

Dense MSE, J_2 , H^1 , maximum error, global signature, 10-block local signature, 100-block local signature and burst-region J_2 .

Numerical audit

Simpson and Romberg quadrature agree beyond any observed loss separation. RK4 refinement changes metrics at a smaller scale still.

Design, analysis and numerical audit: Notebook 05, Sections 9, 17, 20 and 22. Quadrature references: [11,12]

Validation checks for Brownian messages

- ▶ Acceptance tests verify shape, finite values, basis dimensions, Brownian scaling, message support and zero-message identity.
- ▶ RoughPy independently verifies original and modified step-two signatures to tolerance 10^{-9} .
- ▶ Main study uses 10,000 supplied streams. Confirmation uses a disjoint 10,000-stream ensemble.
- ▶ All 29 valid main detection transitions reappear in confirmation. Twenty-three use the same amplitude, four move by one tested amplitude, and two response-derived thresholds move by two.
- ▶ A main raw-area null control failed and passed on confirmation. Main thresholds involving that feature are withheld.

Interpretation rule

Confirmation establishes repeatability of threshold order on another finite ensemble. It does not create a universal detection boundary.

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Project record

Full project bibliography: papers/references.bib. Experiment definitions, stored result paths and analysis code appear in notebooks 01, 05, 06 and 07.